

I. Multiple Choice Questions (MCQ)

1. _____ refers to software in which the source code is freely available for others to view, amend and adapt.
 - a. Free software
 - b. Open source
 - c. Freeware
 - Explanation: The specific definition of "open source" centers on the accessibility of the source code for public viewing and modification.
2. _____ the software is freely accessible and can be freely used, changed, improved, copied and distributed by all who wish to do so. And no payments are needed.
 - a. Free software
 - b. Open source
 - c. Freeware
 - Explanation: "Free software" is defined by the "four freedoms" granted to users, including the right to use and distribute it without cost.
3. _____ can be downloaded from the internet for various purposes, but may not come with the source code.
 - a. Free software
 - b. Open source
 - c. Freeware
 - Explanation: Unlike free software, "freeware" is free in terms of price but often keeps its source code closed, preventing users from modifying it.
4. _____ is software made available with the right to redistribute copies, but it is stipulated that if one intends to use it after a certain period, a license fee should be paid.
 - a. Free software
 - b. Open source
 - c. Shareware
 - Explanation: "Shareware" uses a "try before you buy" model where redistribution is allowed, but ongoing use eventually requires a payment.
5. _____ is a non-profit corporation dedicated to managing and promoting the Open Source definition by certifying products.
 - a. FLOSS
 - b. GNU
 - c. OSI (Open Source Initiative)
 - Explanation: The OSI is the official organization responsible for certifying whether a software product meets the standards of the Open Source definition.

6. _____ is the software that is neither open nor freely available. Its use is regulated and modification is either forbidden or requires special permission.
- a. GNU
 - b. FSF
 - c. Proprietary S/W
 - Explanation: Proprietary software is the opposite of open source; it is closed, exclusive, and its use is strictly controlled by the supplier.
7. _____ It promotes the development and use of free software. It is a non-profit organization.
- a. FSF (Free Software Foundation)
 - b. GNU
 - c. W3C
 - Explanation: The FSF is the non-profit entity that advocates specifically for the advancement of "free software" principles.
8. A _____ is the contract between the software owner and the licensee defining terms of use of software.
- a. Software license
 - b. Contract
 - c. Consideration
 - Explanation: A software license is a legal contract that specifies what rights are granted to the user and what restrictions apply to the software's use.
9. _____ is the opposite of copyright.
- a. Copyleft
 - b. Proprietary
 - c. Closed source
 - Explanation: "Copyleft" flips the traditional use of copyright; instead of preventing distribution, it uses legal rules to ensure that the software remains open for everyone.
10. _____ it can be licensed under other licenses, even closed-source licenses.
- a. Strong copyleft
 - b. Weak copyleft
 - Explanation: "Weak copyleft" is less restrictive, allowing the open-source code to be linked to or included in works that might have closed-source licenses.
11. GPL is _____ copyleft and very _____ license.
- a. Strong, strict
 - b. Strong, permissive
 - c. Weak, strict
 - Explanation: The GPL is "strong" because it ensures derivative works stay open, and it is "strict" because it cannot be mixed with closed-source software.

12. MIT is a _____ license.

- a. Permissive
- b. Strict
- c. Strong
- Explanation: The MIT license is "permissive" because it allows users to do almost anything with the code as long as the original copyright notice is maintained.

13. _____ systems—including GIT—record changes to a file or set of files over time so that you can recall specific versions later.

- a. Operating
- b. Version control
- c. Video
- Explanation: Version control is the technical process of tracking every change made to a project so that you can revert to any previous state at any time.

14. In Git, the term _____ is referring to the remote repository.

- a. Start
- b. origin
- c. Clone
- Explanation: "origin" is the standard alias or shorthand name used by Git to refer to the remote server from which a project was originally cloned.

15. The git _____ command creates a new Git repository.

- a. init
- b. clone
- c. add
- Explanation: The `init` command is used to transform a regular, untracked folder into a fully functioning Git repository.

16. The git _____ command adds a change in the working directory to the staging area.

- a. ignore
- b. branch
- c. add
- Explanation: `git add` is the "staging" step where you select which specific file modifications you want to include in your next saved version (commit).

17. _____ is a lightweight markup language with simple syntax for creating formatted text for README git files.

- a. MPL
- b. Markdown
- c. ogg
- Explanation: Markdown allows developers to write in plain text using simple symbols (like `#` or `*`) that then render as professionally formatted text on platforms like GitHub.

II. True or False Questions

1. FREE SOFTWARE means the software is freely accessible and can be freely used.
 - True.
 - Explanation: This is the core definition of Free Software provided in the sources.
2. In Free Software, the user has the freedom to study how the program works.
 - True.
 - Explanation: This refers to "Freedom 2" of the "Four Freedoms," which explicitly grants the right to study and change the code.
3. Free software is the S/W which permits redistribution but not modification.
 - False.
 - Explanation: Free software must allow modification; the source states that software permitting redistribution but *not* modification is "freeware".
4. One of the properties of open source is free redistribution of software.
 - True.
 - Explanation: Open-source licenses must allow the software to be given away or even sold as part of a package without paying royalties.
5. In a GPL license, licensees can modify the work.
 - True.
 - Explanation: The GPL allows users to modify the original work to create "derivative works" as long as those works are also released under the GPL.
6. Linux is a Closed Source Operating System.
 - False.
 - Explanation: Linux is one of the most prominent examples of an "open-source" operating system kernel.
7. GitHub is a cloud-based hosting service that lets you manage GIT repositories.
 - True.
 - Explanation: GitHub is the web service used to host, share, and collaborate on Git projects online.
8. When using Weak copyleft, the derived work can be licensed under another license or even closed-source licenses.
 - True.
 - Explanation: Weak copyleft is specifically designed to allow open-source files to be part of a project that uses different, potentially proprietary, licenses.

9. The command `git clone <repository>` creates a copy of the repository in the current directory.
- True.
 - Explanation: Cloning is the process of copying an existing repository from a remote server to your local system.
10. In Proprietary Standards, users have to (buy) a license to use them.
- True.
 - Explanation: Proprietary or "closed" standards are owned by companies (like Microsoft or Apple) and typically require a paid license fee.
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III. Short Answer / Essay Questions

1. List 3 types of version control systems.
- Answer: RCS, CVS, Subversion, and Git.
 - Explanation: These are the primary systems that have been used over the years to manage and track code changes.
2. List the "Four Freedoms" provided to users of Free Software.
- Answer: 1. Freedom to run for any purpose; 2. Freedom to study and change; 3. Freedom to redistribute copies; 4. Freedom to improve and release improvements.
 - Explanation: These four pillars define the ethical and legal rights of every Free Software user.
3. What is the difference between Git and GitHub?
- Answer: Git is local software (a command-line tool) used for version control. GitHub is a cloud-based hosting service with a graphical interface used to manage Git repositories.
 - Explanation: Git is the engine that does the work, while GitHub is the website where you share that work with others.
4. Explain the basic Git workflow.
- Answer: 1. Modify files in the working directory; 2. Stage the changes (`git add`); 3. Commit the changes to the local repository history.
 - Explanation: This three-step process ensures that changes are reviewed (staged) before they are permanently saved (committed) to the project history.
5. Name three examples of FLOSS (Free/Libre/Open Source Software) projects.
- Answer: Linux, Ubuntu, Android, GIMP, Blender, Mozilla Firefox, or LibreOffice.
 - Explanation: These projects are widely used and represent the core of the open-source software ecosystem.

